OPENER: Open Partitioning Embedding for Named Entity Recognition

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Abstract

Open-world named entity recognition (NER) must assign entity types that were not fixed at training time. Recent systems either train a dedicated encoder or prompt billion-parameter language models, and they are compared almost exclusively on accuracy, ignoring their compute and energy cost. We present **OPENER**, a lightweight open-world NER pipeline assembled from off-the-shelf components: a frozen mention detector, a Matryoshka-truncatable sentence embedder sharpened by a light contrastive fine-tuning, and a balanced linear classifier for entity typing, so that no encoder is trained from scratch. We benchmark OPENER against GLiNER, GNER and OWNER on 13 datasets along three axes at once: quality, measured by Adjusted Mutual Information (AMI), inference latency, and energy. Our analysis separates entity typing on gold mentions, where the contrastive embedding space is highly discriminative, from the end-to-end setting, where the dominant open-world bottleneck is shown to be mention detection rather than typing. The energy-aware comparison further exposes order-of-magnitude efficiency gaps between approaches that a quality-only evaluation would hide. On gold mentions, OPENER's balanced linear typing surpasses a zero-shot transfer of OWNER (54.0 vs 43.0 AMI), while end-to-end the systems fall within a few AMI points of one another as mention detection becomes the shared bottleneck; OPENER reaches this quality at $3.4\,$Wh per run, one to two orders of magnitude below the generative ($36.5\,$Wh) and contrastive-clustering ($13.6\,$Wh) baselines. We release the code, the per-experiment configurations and the full benchmark.

Index Terms

named entity recognition, open NER, Matryoshka embeddings, Gaussian mixture models, semi-supervised clustering, out-of-distribution detection

I. Introduction

[Placeholder — introduction to be written.]

II. Related Work

OPENER sits at the intersection of three lines of research: open-world and zero-shot named entity recognition (NER), pretrained entity representations refined by contrastive learning, and energy-aware machine learning. We review each in turn and state how OPENER differs.

A. Closed-set and Span-based NER

Traditional NER is cast as supervised sequence labeling over a fixed inventory of entity types, typically with a pretrained encoder followed by a token- or span-level classifier. Span-based formulations enumerate candidate spans and classify them directly, which simplifies the handling of nested and discontinuous entities [1]. These models reach high accuracy on in-domain benchmarks such as CoNLL-2003 [2], but their label set is frozen at training time. Adding a new entity type requires re-annotating data and retraining the model, which is the central limitation that open-world NER seeks to remove.

B. Open-world and Zero-shot NER

Open-world NER aims to recognise entity types that were not fixed at training time. GLiNER [3] treats the candidate type names as queries to a bidirectional encoder, a DeBERTaV3 backbone [4], and scores every span against them in a single forward pass, offering a fast alternative to autoregressive tagging. GNER [5] instead casts NER as text generation with a sequence-to-sequence model [6] and shows that explicitly modeling negative, non-entity instances sharpens type boundaries. Closest to our work, Genest et al. [7] propose OWNER, an unsupervised open-world pipeline that decouples mention detection from entity typing: it embeds detected mentions with an encoder refined by contrastive learning and then clusters them into dynamically named types. Other approaches prompt a general-purpose large language model to perform extraction directly, as in ChatIE [8], or focus on recovering entities that detectors tend to miss [9]. OPENER adopts the decoupled detect-then-type view of OWNER, but replaces the trained encoder and the unsupervised clustering with a pretrained, Matryoshka-truncatable embedding space and a balanced linear classifier, so that no encoder is trained from scratch.

C. LLM-based and Few-shot NER

A complementary line distils or instruction-tunes large language models for open NER. UniversalNER [10] distils a chat model into a smaller LLaMA-based student [11] over tens of thousands of entity types, while GoLLIE [12] conditions a code language model on annotation guidelines to follow unseen schemas. Few-shot methods reduce supervision further through prompt optimization [13], knowledge-graph-guided contrastive learning [14], or token-level contrastive objectives such as CONTaiNER [15]. These methods are accurate but rest on models with billions of parameters, whose inference cost, latency and energy use are substantial, and which may not even fit on commodity hardware. This cost is precisely what motivates the lightweight design and the energy-aware evaluation of OPENER.

D. Entity Representations and Contrastive Learning

OPENER relies on a pretrained sentence embedder rather than a task-specific encoder. Nomic Embed [16] provides an open, long-context model trained with the Matryoshka objective [17], whose representations can be truncated to lower dimensions without retraining, trading quality for compute. Such embedders build on the Sentence-BERT line of work [18], and we sharpen the entity-type geometry with a triplet margin objective in the spirit of FaceNet [19]. Embedding labels and entities in a shared space [20], modeling entity types with Gaussian components [21], generating synthetic entity-bearing text for augmentation [22], and using mixtures to augment training data [23] are recurring ideas in low-resource NER, all grounded in classical Gaussian mixture models fitted by expectation-maximization [24], [25]. Unlike methods that train a dedicated encoder, OPENER keeps the pretrained space largely frozen and only applies a light contrastive fine-tuning, which tests whether a general-purpose embedding is already discriminative enough for entity typing.

E. Efficient and Green AI

A growing body of work argues that efficiency should be reported alongside accuracy. Strubell et al. [26] quantify the financial and environmental cost of training NLP models, Schwartz et al. [27] advocate efficiency as a first-class evaluation criterion, and Patterson et al. [28] analyse the carbon footprint of machine-learning training and how to reduce it. Tools such as CodeCarbon [29] make per-run energy and carbon estimates practical. Open-world NER systems, however, are still compared almost exclusively on quality. OPENER instead reports a three-axis comparison of quality, measured by Adjusted Mutual Information [30], inference speed, and energy, placing the frugality of each method on an equal footing with its accuracy.

In summary, existing open-world NER methods either train a dedicated encoder [7] or query costly large language models [10], [12], and they are compared on accuracy alone. OPENER fills both gaps: it reuses pretrained components with only a light contrastive step and a balanced linear classifier, and it is evaluated jointly on quality, speed and energy.

III. Method

[Placeholder — method to be written.]

IV. Experiments

A. Datasets

We evaluate on **13 evaluation sets** that span news, social media, spoken queries, encyclopedic text, biomedical abstracts and manufacturing literature, following the dataset selection of Genest et al. [7]. CrossNER [31] contributes five domain-specific sets, evaluated separately (artificial intelligence, literature, music, politics, science). The remaining eight are CoNLL-2003 [2], WNUT-17 [32], the MIT-Restaurant and MIT-Movie spoken-query corpora [33], FabNER [34], the BioNLP/JNLPBA-2004 biomedical task [35], and the GUM [36] and GENTLE [37] multi-genre corpora. Label granularity ranges from four coarse types in CoNLL-2003 to twelve fine-grained categories in FabNER, including opaque domain acronyms. Two datasets used by [7], GENIA [38] and i2b2, are license-gated and therefore not re-measured here; BioNLP/JNLPBA-2004 serves as our biomedical proxy. To homogenise evaluation cost across systems, we cap each test split at 1000 sentences (the full split is used when it is smaller). Table I summarises the benchmark.

B. Baselines

We compare OPENER against representative open-world NER systems run end-to-end on the same splits. GLiNER [3] is evaluated in its small, medium and large variants to trace a size/efficiency trade-off. GNER [5] is a generative tagger built on T5 [6]; we run the T5-base checkpoint. We further compare against OWNER [7], executed in its original environment. Instruction-tuned and few-shot large-language-model NER systems, namely UniversalNER [10], GoLLIE [12] and ChatIE [8], rely on 7B-parameter backbones [11] that do not fit our hardware budget even under 4-bit NF4 quantization [39]; for these we report the figures published by their authors rather than re-measuring them.

TABLE I
THE 13 EVALUATION SETS. "TEST" IS THE NUMBER OF TEST SENTENCES USED (CAPPED AT 1000).

Dataset	Abbr.	Domain	Test
CrossNER-AI	AI	Encyclopedic (AI)	430
CrossNER-Literature	Lit	Encyclopedic (literature)	416
CrossNER-Music	Mus	Encyclopedic (music)	465
CrossNER-Politics	Pol	Encyclopedic (politics)	651
CrossNER-Science	Sci	Encyclopedic (science)	543
WNUT-17	WNUT	Social media	689
MIT-Restaurant	Rest	Restaurant search	1000
MIT-Movie	Movie	Movie trivia	1000
FabNER	Fab	Manufacturing	1000
BioNLP/JNLPBA-2004	Bio	Biomedical	1000
CoNLL-2003	CoNLL	News	1000
GUM	GUM	Multi-genre	1000
GENTLE	GEN	Multi-genre (OOD)	1000

C. Metrics

Our primary metric is the Adjusted Mutual Information (AMI) [30] between predicted and gold type assignments, following [7], complemented by macro-averaged F_1 and accuracy. Because open-world predictions need not share the gold label vocabulary, AMI is computed over the aligned set of mentions, which makes it well suited to comparing dynamically named types. Beyond quality, we report two efficiency axes: inference *speed*, as the median (p50) latency per sentence, and *energy*, as kilowatt-hours and grams of CO_2eq per run.

D. Evaluation Protocol

Predicted and gold mentions are aligned by exact character offsets. Following [7], an unmatched gold mention (a false negative) and an unmatched predicted mention (a false positive) are each assigned a distinct sentinel label before computing AMI, so that detection errors are penalised rather than ignored. We distinguish two protocols that must not be mixed in the same comparison. In the *typing-on-gold* protocol, the entity-typing component is applied to gold mention spans, which measures typing quality in isolation under perfect detection. In the *end-to-end* protocol, mentions are produced by the detector and the sentinel labels above are active, which is the setting comparable to the baselines. All runs use a single fixed random seed (42); we note this as a limitation, since we do not report variance over seeds.

E. Implementation Details

All experiments run on a single consumer-grade GPU with 6 GB of VRAM (CUDA 12.1, PyTorch 2.5.1 [40]). We use the HuggingFace Transformers library [41] for the baselines, sentence-transformers [18] for the Matryoshka embedder [16], [17], and scikit-learn [42] for the linear classifier, whose solver is LIBLINEAR [43]. Latency is measured with two warmup passes followed by timed passes with GPU synchronisation, reporting p50, p95 and p99 per sentence. Energy is measured with CodeCarbon [29] in offline mode using the French grid carbon intensity; when hardware power counters are unavailable, we fall back to a thermal-design-power estimate and flag the measurement method in the report. To keep the energy comparison fair, all systems are measured on the same machine within a single session, under comparable thermal conditions.

F. Main Results

The tables below report the benchmark on all 13 sets. OWNER is run in its native protocol [7]: a single model is trained once on the CoNLL-2003 source (a one-time cost of 88 minutes on our 6 GB GPU) and then applied *zero-shot* to every target set (cells marked §). CoNLL-2003 is therefore in-domain for OWNER (marked †) and is excluded when reading OWNER as a transfer system. One protocol asymmetry must be kept in mind when reading the typing scores: OPENER fits a lightweight linear classifier on each target's training split (a supervised probe), whereas OWNER, GLiNER and GNER use no target-side labels. The two systems thus trade annotation for inference cost: OPENER needs target labels but stays cheap at test time, while OWNER needs none but pays a heavy per-sentence cost (Table III).

G. Ablations

We justify each architectural choice with a one-factor-at-a-time search. Starting from the default pipeline, we vary a single component while holding the others fixed, and keep the setting with the highest mean AMI. Table VI reports this search; the protocol comparison between typing-on-gold and end-to-end is discussed alongside the main results. To isolate the entity-typing component itself, Table VII compares OPENER against OWNER on gold mentions, where detection errors are removed. *[Some axes are still being re-measured under the current pipeline; see the notes in Table VI.]*

TABLE II

Open-world NER, end-to-end protocol. Adjusted Mutual Information (AMI $\times 100$, $\uparrow$). Abbreviations follow Table I; bold marks the best system per column. § OWNER is zero-shot (trained once on CoNLL-2003, applied to each target). † CoNLL-2003 is in-domain for OWNER (its training source) and is not bolded as a transfer result; the best zero-shot system on CoNLL is GNER. The typing-on-gold comparison is reported in Table VII.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	41.7	49.8	56.4	50.6	53.3	29.7	26.1	35.6	10.7	31.8	18.9	**34.1**	**36.1**	36.5
GLiNER-M	42.0	**50.1**	57.4	50.2	**54.2**	28.1	27.9	36.4	11.3	**34.9**	24.9	32.8	31.6	37.1
GLiNER-L	42.9	48.9	**58.0**	49.8	52.8	28.9	**30.9**	**37.6**	28.1	34.6	29.7	32.7	30.5	**38.9**
GNER	41.9	46.9	53.5	48.9	49.9	**30.0**	30.0	36.8	17.0	29.1	**43.7**	32.8	31.6	37.9
OWNER§	**43.9**	44.0	46.6	**50.9**	50.3	22.3	7.6	24.0	18.7	26.5	50.3†	29.1	31.3	34.3
OPENER	39.1	39.3	49.2	44.2	47.8	24.8	30.6	36.7	**28.3**	31.4	39.4	32.5	32.5	36.6

TABLE III

Inference latency. Median per-sentence latency in milliseconds ($\downarrow$), end-to-end. Abbreviations follow Table I; best per column in bold. § OWNER does not expose a per-sentence call: its figure is the total inference wall-clock divided by the number of sentences (an amortised mean, not a true p50), and excludes the one-time 88-min training on CoNLL-2003. CoNLL († in-domain) is omitted as its timing is folded into that training run. OWNER is an order of magnitude slower than GLiNER because it re-encodes every mention and clusters at inference time.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	**21**	**22**	**21**	**21**	**21**	**20**	**19**	**20**	**20**	**22**	**20**	**21**	**21**	**21**
GLiNER-M	36	36	37	38	39	39	38	40	41	42	44	46	45	40
GLiNER-L	144	149	153	149	155	137	132	145	147	143	131	141	137	143
GNER	3431	5408	7461	6472	7110	2906	1610	3344	4310	5034	2914	3435	1500	4226
OWNER§	500	712	895	920	757	311	281	532	438	558	–	759	626	607
OPENER	105	519	247	153	227	103	97	111	96	119	100	104	101	160

TABLE IV

Energy. Energy per run in watt-hours ($\downarrow$), end-to-end. Abbreviations follow Table I; best per column in bold. § OWNER is measured at inference only (zero-shot transfer); its one-time CoNLL-2003 training (88 min) is amortised across all targets and not charged per run. CoNLL (†) is omitted as its energy is folded into that training. OWNER remains energy-heavy because each sentence triggers per-mention encoding and clustering.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	**0.19**	**0.17**	**0.20**	**0.27**	**0.22**	**0.27**	**0.35**	**0.39**	**0.40**	**0.42**	**0.37**	**0.41**	**0.40**	**0.31**
GLiNER-M	0.31	0.29	0.33	0.48	0.40	0.50	0.66	0.76	0.77	0.79	0.69	0.82	0.75	0.58
GLiNER-L	0.68	0.67	0.77	1.04	0.91	1.01	1.37	1.51	1.54	1.49	1.38	1.52	1.43	1.18
GNER	17.13	24.28	39.75	50.88	44.76	20.60	17.02	37.17	49.11	63.50	45.22	38.69	26.50	36.51
OWNER§	6.6	9.0	12.7	18.3	12.6	6.5	8.6	16.3	13.4	17.1	–	23.2	19.1	13.6
OPENER	0.86	2.16	22.76	1.89	2.21	1.27	1.66	2.06	1.66	2.08	1.70	1.93	1.82	3.39

TABLE V

Summary: quality versus efficiency, averaged over the 13 datasets. AMI ($\times 100$, $\uparrow$); median per-sentence latency (ms), energy (Wh) and carbon (gCO_2eq) per run ($\downarrow$); and parameter count. § OWNER is zero-shot (trained once on CoNLL-2003 for 88 min, then applied to every target); its latency is an amortised wall-clock mean (Table III); latency and energy are averaged over the 12 transfer sets (the in-domain CoNLL cost is folded into training), while AMI is the 13-set mean of Table II. End-to-end OPENER pairs the GLiNER-L detector (330M) with the 137M embedder, hence 467M; on gold mentions the detector is not used (137M). End-to-end OWNER likewise sums a DeBERTa-v3-base detector ($\approx$184M) and a BERT-base typing encoder (110M), hence $\approx$294M; GLiNER-S/M/L sizes are from the project notes and unverified.

Model	Params	AMI	p50 (ms)	Energy (Wh)	CO_2 (g)
GLiNER-S	50M	36.5	21	0.31	0.02
GLiNER-M	200M	37.1	40	0.58	0.03
GLiNER-L	330M	38.9	143	1.18	0.07
GNER	220M	37.9	4226	36.51	2.05
OWNER§	294M	34.3	607	13.6	0.71
OPENER	467M	36.6	160	3.39	0.19

TABLE VI

OPENER ARCHITECTURE SEARCH. ONE COMPONENT IS VARIED AT A TIME AROUND THE DEFAULT PIPELINE (**BOLD**). AMI AND MACRO-F_1 ($\times 100, \uparrow$); PER-RUN LATENCY, ENERGY AND CARBON ($\downarrow$). ‡ ENTITY-TYPING SWEEP ON A 10-DATASET SUBSET WITH THE CONTRASTIVE EMBEDDER; § DIMENSION SWEEP ON A 6-DATASET PILOT WITH THE FROZEN EMBEDDER. LATENCY AND ENERGY ARE DOMINATED BY THE EMBEDDER AND DETECTOR, HENCE NEAR-CONSTANT ACROSS CLASSIFIERS; PER-AXIS EFFICIENCY FOR THE CURRENT PIPELINE IS BEING RE-MEASURED.

Configuration	AMI	F_1	p50 (ms)	En. (Wh)	CO_2 (g)
Mention detector					
GLiNER-S	–	–	–	–	–
GLiNER-M	–	–	–	–	–
GLiNER-L	–	–	–	–	–
Entity embedder					
frozen Nomic v1.5	–	–	–	–	–
+ contrastive fine-tuning	–	–	–	–	–
Matryoshka dimension§					
64	15.1	–	–	–	–
128	15.4	–	–	–	–
256	17.7	–	–	–	–
512	18.3	–	–	–	–
768	**18.8**	–	–	–	–
Entity typing‡					
GMM (anchor warm-start)	–	–	–	–	–
LogReg	39.2	30.6	–	–	–
LogReg (balanced)	49.4	55.2	–	–	–
LinearSVC	52.4	50.2	–	–	–
LinearSVC (balanced)	**56.6**	**64.5**	100	3.32	0.18

TABLE VII

ENTITY TYPING ON GOLD MENTIONS: OPENER VS OWNER (AMI $\times 100, \uparrow$). DETECTION IS BYPASSED, ISOLATING THE TYPING COMPONENT. ABBREVIATIONS FOLLOW TABLE I; BOLD MARKS THE PER-DATASET WINNER. THE TWO SYSTEMS USE DIFFERENT SUPERVISION: OPENER FITS A LINEAR PROBE ON EACH TARGET'S TRAINING LABELS, WHEREAS OWNER IS ZERO-SHOT (CLUSTERING AFTER TRANSFER FROM CONLL-2003, §). UNDER THIS PROTOCOL OPENER LEADS ON 11 OF 13 SETS. † CONLL-2003 IS IN-DOMAIN FOR OWNER (ITS TRAINING SOURCE); IT IS NOT A TRANSFER RESULT AND IS NOT BOLDED.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
OWNER§	**55.7**	**54.3**	59.8	63.2	63.1	30.5	29.3	31.8	31.9	29.2	52.5†	30.0	27.2	43.0
OPENER	55.6	53.5	**64.2**	**66.6**	**66.1**	**41.8**	**51.2**	**54.2**	**36.5**	**60.2**	**70.7**	**37.7**	**44.2**	**54.0**

V. CONCLUSION

[Placeholder — conclusion to be written.]

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[Placeholder — to be filled at submission time.]

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